NANYANG TECHNOLOGICAL UNIVERSITY

SEMESTER 1 EXAMINATION 2025-2026

EE6222 – MACHINE VISION

November / December 2025 Time Allowed: 3 hours

INSTRUCTIONS

1. This paper contains 4 questions and comprises 3 pages.
2. Answer all 4 questions.
3. All questions carry equal marks.
4. This is a closed-book examination.
5. Unless specifically stated, all symbols have their usual meanings.

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1. A digital image of size 3×3 quantized by 2 bits is plotted below:

1	2	0
1	3	1
0	2	1

(a) Express or plot the histogram $h_f(f)$ of the image $f(x, y)$.

(5 Marks)

(b) A power transform $T(f) = \frac{1}{3}f^2$ is applied to $f(x, y)$ to generate image $g(x, y)$ quantized by 2 bits. Express or plot the generated image $g(x, y)$ and its histogram $h_g(g)$.

(10 Marks)

(c) Generate a histogram equalized image $w(x, y)$ from $f(x, y)$ and quantize it by 2 bits. Express or plot your generated image $w(x, y)$ and its histogram $h_w(w)$.

(10 Marks)

2. Reducing high-dimensional data $\mathbf{x}$ to low dimensional data $\mathbf{y}$ can be performed by linear transform $\mathbf{y} = \mathbf{W}^T\mathbf{x}$, where $\mathbf{x}$ and $\mathbf{y}$ are column vectors and $\mathbf{W}$ is a matrix.

   (a) Express the reconstructed high-dimensional data $\hat{\mathbf{x}}$ from the reduced low-dimensional data $\mathbf{y}$ and express the reconstruction error $e^2 = (\hat{\mathbf{x}} - \mathbf{x})^T(\hat{\mathbf{x}} - \mathbf{x})$ in terms $\mathbf{x}$ of and $\mathbf{W}$.

   (7 Marks)

   (b) Three-dimensional data can be represented by $\mathbf{x} = (x_1 \quad x_2 \quad x_3)^T$. Obtain the transform matrix $\mathbf{W}$ if $\mathbf{y}$ captures the second and third components of $\mathbf{x}$, i.e., $\mathbf{y} = (x_2 \quad x_3)^T$, and compute the reconstruction error.

   (8 Marks)

   (c) Two data samples $\mathbf{x}_1 = (1 \quad -\sqrt{2} \quad -1)^T$ and $\mathbf{x}_2 = (-3 \quad 3\sqrt{2} \quad 3)^T$ are reduced to one dimension by transform matrix $\mathbf{W} = (0.5 \quad -0.5\sqrt{2} \quad -0.5)^T$. Compute the reduced one dimensional samples $\mathbf{y}_1$, $\mathbf{y}_2$, the reconstructed samples $\hat{\mathbf{x}}_1$, $\hat{\mathbf{x}}_2$ and reconstruction errors ${e_1}^2$ and ${e_2}^2$.

   (10 Marks)

3. A 3×3 convolutional kernel of CNN learnt by the training data is shown below:

−1	1	−1
1	1	1
−1	1	−1

A 6×6 digital image is shown below:

2	2	2	0	2	0
2	2	2	0	2	0
2	2	2	0	0	0
0	0	0	0	2	0
2	2	0	2	2	2
2	2	0	0	2	0

Answer the following questions:

   (a) Assume that the learnt bias is zero. Plot the feature maps of size 4×4 before and after the Relu activation function.

   (8 Marks)

   (b) Assume that the learnt bias is −5. Plot the feature maps of size 4×4 after the Relu activation function and describe an important feature of CNN.

   (5 Marks)

Note: Question No. 3 continues on page 3.

(c) The attention mechanism of Transformer generates 3 matrices $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ from input matrix $\mathbf{X}$ by $\mathbf{Q} = \mathbf{X}\mathbf{W}_q$, $\mathbf{K} = \mathbf{X}\mathbf{W}_k$, $\mathbf{V} = \mathbf{X}\mathbf{W}_v$, which project input tokens $\mathbf{x}_i$ into $\mathbf{q}_i$, $\mathbf{k}_i$, $\mathbf{v}_i$.

(i) How to derive $\mathbf{q}_i$, $\mathbf{k}_i$, $\mathbf{v}_i$ from matrices $\mathbf{Q}$, $\mathbf{K}$, $\mathbf{V}$ and express $\mathbf{q}_i$, $\mathbf{k}_i$, $\mathbf{v}_i$ in terms of $\mathbf{x}_i$.

(ii) The output matrix $\mathbf{Y}$ that contains all output tokens $\mathbf{y}_k$ is generated by $\mathbf{Y} = \mathbf{Q}\mathbf{K}^T\mathbf{V}$ (ignoring the scaling factor). Express the output token $\mathbf{y}_k$ in terms of $\mathbf{q}_i$, $\mathbf{k}_i$, $\mathbf{v}_i$ and elaborate the meanings or roles of the Transformer attention mechanism.

(12 Marks)

4. Deep learning has been used for keypoint matching and 3D reconstruction.

(a) Compare the principles of Scale-Invariant Feature Transform and SuperPoint for feature detection and description. What are the key differences in how they identify features and the key differences in how they represent features?

(10 Marks)

(b) Stereo matching is a key step in reconstructing 3D structure from two images.

(i) Explain the steps of stereo matching.

(ii) One of the major challenges in stereo matching is how to deal with textureless regions, occlusions, and repetitive patterns. Discuss how these challenges affect disparity estimation and describe one strategy to mitigate each of them in the learning-based method.

(15 Marks)

END OF PAPER

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Please read the following instructions carefully:

1. **Please do not turn over the question paper until you are told to do so. Disciplinary action may be taken against you if you do so.**

2. You are not allowed to leave the examination hall unless accompanied by an invigilator. You may raise your hand if you need to communicate with the invigilator.

3. Please write your Matriculation Number on the front of the answer book.

4. Please indicate clearly in the answer book (at the appropriate place) if you are continuing the answer to a question elsewhere in the book.